

NAM DINH, Viet Nam

WMO: 488230

Lat: 20.4333N

Lon: 106.1570E

Elev: 10

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	49.8	51.7	39.4	35.5	57.3	42.6	40.2	56.3	14.8	61.2	13.6	61.0	5.7	340	0.294

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	10.7	96.2	81.9	94.4	81.9	92.6	81.7	83.8	90.6	83.3	90.0	82.7	89.2	5.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
82.3	168.6	87.1	81.7	165.0	86.7	81.1	161.7	86.3	48.2	91.0	47.4	90.4	46.9	89.5	92.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
13.1	11.2	9.7	DB	47.0	99.7	2.8	1.9	45.0	101.0	43.4	102.1	41.8	103.2	39.8	104.6
			WB	44.4	85.9	2.7	1.5	42.5	87.0	40.9	87.9	39.3	88.7	37.4	89.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	76.1	62.8	65.0	69.1	76.6	82.5	86.3	85.9	84.4	82.3	78.8	72.7	65.6	(6)
(7)		DBStd	9.62	6.51	6.96	5.62	5.08	4.27	3.36	3.23	2.99	3.31	4.03	5.31	5.79	(7)
(8)		HDD50	2	1	1	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	299	123	81	25	2	0	0	0	0	0	0	6	62	(9)
(10)		CDD50	9513	396	422	593	797	1009	1088	1114	1065	970	893	682	484	(10)
(11)		CDD65	4336	54	82	153	349	544	638	649	600	520	428	238	81	(11)
(12)		CDH74	44094	70	177	434	2439	6038	8378	8469	7279	5654	3604	1317	235	(12)
(13)		CDH80	17621	1	17	41	528	2411	4198	4121	3049	2019	986	232	18	(13)
(14)	Wind	WSAvg	4.4	4.4	4.3	4.1	4.6	4.9	4.8	4.8	4.0	4.1	4.2	4.2	4.1	(14)
(15)	Precipitation	PrecAvg	65.0	1.1	0.9	2.0	2.6	6.9	7.1	9.7	12.0	12.5	6.3	2.7	1.1	(15)
(16)		PrecMax	109.3	6.9	2.0	5.0	8.0	14.8	16.8	16.3	21.6	27.8	20.0	9.3	3.7	(16)
(17)		PrecMin	45.9	0.0	0.2	0.2	0.8	2.5	3.5	3.3	5.2	4.7	1.1	0.2	0.1	(17)
(18)		PrecStd	12.3	1.5	0.5	1.4	1.9	3.0	2.8	3.1	4.2	5.4	4.5	2.4	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	78.8	81.9	84.3	92.3	97.2	99.0	98.3	96.0	93.5	90.2	87.2	82.4	(19)	
(20)		MCWB	71.4	74.1	76.3	81.1	82.3	81.4	82.0	82.2	81.5	77.7	77.2	74.2	(20)	
(21)		DB	75.9	78.2	80.5	87.5	93.9	96.7	95.5	93.6	91.4	88.5	84.9	79.2	(21)	
(22)		MCWB	69.9	72.4	74.6	79.5	82.3	81.7	82.0	82.3	80.9	77.2	75.9	72.2	(22)	
(23)		DB	73.7	76.0	78.1	85.2	91.2	94.4	93.5	91.7	89.9	87.1	82.8	76.5	(23)	
(24)		MCWB	68.6	71.5	73.8	78.5	81.6	81.7	81.7	81.9	80.3	76.8	74.2	70.9	(24)	
(25)		DB	71.6	74.1	76.3	82.9	89.0	92.3	91.8	89.9	88.2	85.4	80.6	74.1	(25)	
(26)		MCWB	67.7	70.7	72.9	77.5	80.8	81.7	81.4	81.5	79.6	76.3	73.2	68.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	73.3	75.0	77.2	82.1	84.3	84.9	84.5	84.5	83.6	80.8	79.0	76.1	(27)	
(28)		MCDWB	76.5	79.1	81.8	89.6	92.5	91.9	92.1	90.6	89.5	86.2	83.5	79.9	(28)	
(29)		WB	71.1	73.5	75.7	80.2	83.1	83.7	83.5	83.5	82.6	79.6	77.8	74.0	(29)	
(30)		MCDWB	74.3	76.7	78.9	85.9	90.4	90.5	90.6	89.5	87.8	84.5	81.7	77.7	(30)	
(31)		WB	69.8	72.4	74.6	79.2	82.3	83.0	82.8	82.8	81.8	78.8	76.7	72.2	(31)	
(32)		MCDWB	72.9	75.3	77.3	83.8	89.1	89.8	89.5	88.8	86.8	83.4	80.5	75.4	(32)	
(33)		WB	68.3	71.2	73.5	78.2	81.6	82.5	82.3	82.2	81.0	78.1	75.5	70.5	(33)	
(34)		MCDWB	71.0	73.8	75.9	82.1	87.6	89.1	88.7	88.0	85.9	82.5	79.6	73.2	(34)	
(35)	Mean Daily Temperature Range	MDBR	8.4	7.6	7.5	8.6	10.4	10.7	9.7	9.4	9.4	10.3	10.5	10.1	(35)	
(36)		5% DB	10.9	10.0	9.4	11.4	13.2	13.1	12.1	12.0	11.6	12.1	12.0	11.9	(36)	
(37)		MCWBR	5.4	4.7	4.8	5.1	5.0	4.0	4.0	4.2	3.6	3.5	4.1	5.3	(37)	
(38)		5% WB	9.3	8.6	8.3	10.7	12.1	11.5	11.0	10.8	10.5	10.5	10.5	10.6	(38)	
(39)		MCWBR	5.5	4.7	4.9	5.4	5.1	4.4	4.3	4.6	4.4	3.9	4.7	5.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.613	0.646	0.809	0.763	0.562	0.513	0.484	0.539	0.599	0.707	0.613	0.643	(40)	
(41)		taud	1.725	1.700	1.450	1.562	2.040	2.197	2.273	2.092	1.895	1.635	1.778	1.693	(41)	
(42)		Ebn at Noon	210	214	187	199	241	251	259	246	230	197	208	196	(42)	
(43)		Edh at Noon	69	75	99	89	55	47	43	52	62	78	65	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	746	867	952	1261	1601	1657	1593	1462	1382	1232	1065	826	(44)	
(45)		RadStd	174	144	115	129	140	132	168	113	101	90	131	126	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.67	N/A	-1.67	+1.08	+0.38	+0.36	N/A	N/A	N/A	N/A	(46)
(47)	Regional (5 neighbors)	N/A	N/A	N/A	+0.99	+0.36	+0.38	N/A	N/A	N/A	+198	(47)

Nomenclature: See separate page